

Exploratory Data Analysis (EDA):

Project Context

This project examines whether shifts in federal policy regimes are associated with structural changes in U.S. immigration enforcement outcomes between 2015 and 2025. Rather than focusing on raw totals, the analysis centers on the “repatriation to encounter” ratio as a measure of enforcement intensity. The goal is not to prove causation, but to evaluate whether observable structural differences in enforcement outcomes coincide with identifiable regime transitions such as administration changes and Title 42 activation.

Numerical and Categorical Features

The DHS repatriations dataset mainly includes annual removal totals by fiscal year, along with several categorical breakdowns such as repatriation type, criminal vs non criminal status, arrest location, citizenship, and agency. For this EDA, the most important numerical variable is the annual repatriation total (quantity), since it represents the outcome side of immigration enforcement. These values are count based, meaning they are whole numbers and cannot be negative, and they show noticeable variation across fiscal years. This variability becomes clear in the Annual Repatriations line plot shown below, where we can see fluctuations over time rather than a stable pattern.

The CBP encounters dataset, on the other hand, represents inflow pressure. Its main numerical variable is Encounter Count, which captures how many individuals were encountered in a given fiscal year. In addition to year level totals, this dataset includes several categorical dimensions such as Land Border Region, Area of Responsibility, State, Demographic category, Citizenship, Title of Authority, and Encounter Type. These categories help describe not just how many encounters occurred, but also under what legal authority and in which geographic or demographic context they occurred. The Annual Encounters line plot below shows a sharp increase beginning around FY2020, highlighting how dramatically inflow pressure changes during this period.

Comparing the two visuals side by side reveals an important pattern. While encounters increase sharply after FY2020, repatriations do not rise at the same rate and eventually decline. This suggests that enforcement outcomes may not scale proportionally with encounter volume. Additionally, the categorical breakdown in the encounters dataset shows a clear distinction between Title 42 and Title 8 authority, which is important given the project’s focus on policy regimes. The Encounters by Title of Authority visualization further supports this by showing how legal authority shifts during peak encounter years, reinforcing the broader regime based framing described in the project proposal.

```
In [105... import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style = "whitegrid")
```

```
In [2]: # DHS Repatriations
dhs = pd.read_excel("KHSM Repatriations fy25m11.xlsx")

# CBP Encounters (combine multiple files)
enc1 = pd.read_excel("nationwide-encounters-fy20-fy23-aor.xlsx")
enc2 = pd.read_excel("nationwide-encounters-fy23-fy26-oct-aor.xlsx")
enc3 = pd.read_excel("nationwide-encounters-fy23-fy26-nov-aor.xlsx")
enc4 = pd.read_excel("nationwide-encounters-fy23-fy26-dec-aor.xlsx")
enc5 = pd.read_excel("nationwide-encounters-fy23-fy26-jan-aor.xlsx")

# Policy Markers
policy = pd.read_csv("policy_markers_2015_2025.csv")
```

```
In [348... dhs.keys()
```

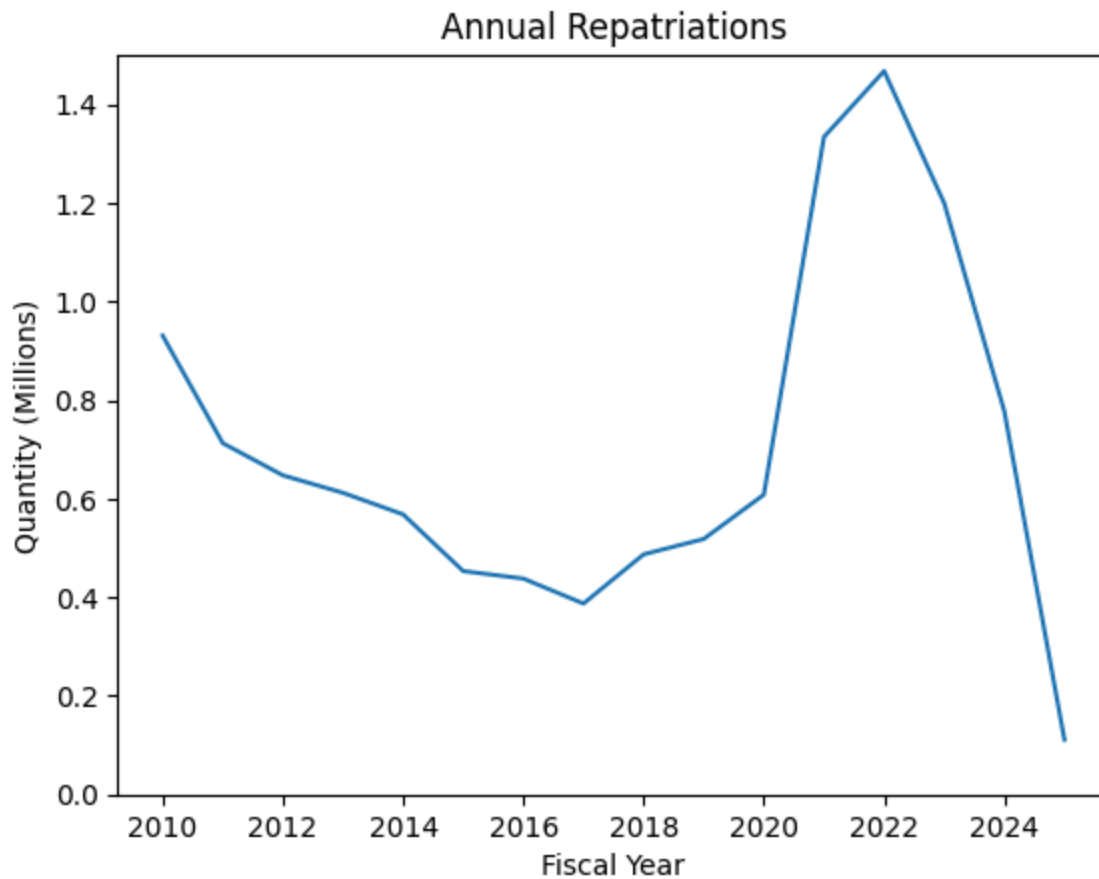
```
Out[348... dict_keys(['Annual Repatriations', 'Monthly Repatriations', 'Annual Repatriation T
ype', 'Monthly Repatriation Type', 'Annual Criminal', 'Monthly Criminal', 'Annual
Arrest Location', 'Monthly Arrest Location', 'Annual Citizenship', 'Monthly Citize
nship', 'Annual Repatriation Agency', 'Monthly Repatriation Agency', 'Annual Type
& Dtl Type', 'Monthly Type & Dtl Type', 'Annual Type & Criminal', 'Monthly Type &
Criminal', 'Annual Type & Arrest Loc', 'Monthly Type & Arr Loc', 'Annual Type & Ci
tp', 'Monthly Type & Citp', 'Annual Typ & Rep Agency', 'Monthly Typ & Rep Agenc
y'])
```

```
In [349... annual_repats = dhs["Annual Repatriations"]
annual_repats.columns
```

```
Out[349... Index(['Fiscal\nYear', 'Quantity'], dtype='object')
```

```
In [350... fig, ax = plt.subplots()
ax.plot(annual_repats["Fiscal\nYear"], annual_repats["Quantity"].apply(lambda x: x/
ax.set_title("Annual Repatriations")
ax.set_xlabel("Fiscal Year")
ax.set_ylabel("Quantity (Millions)")
ax.set_ylim((0,1.5))
```

```
Out[350... (0.0, 1.5)
```



```
In [100...] encounters2023 = pd.read_csv("nationwide-encounters-fy20-fy23-state.csv")
encounters2124 = pd.read_csv("nationwide-encounters-fy21-fy24-state.csv")
encounters2526 = pd.read_csv("nationwide-encounters-fy23-fy26-jan-state.csv")
encounters2526 = encounters2526[encounters2526["Fiscal Year"].apply(lambda x: ("202
encounters2526["Fiscal Year"] = encounters2526["Fiscal Year"].apply(lambda x: 2025
```

```
In [101...] encounters = pd.concat([encounters2023, encounters2124, encounters2526], axis=0)
encounters.columns
```

```
Out[101...] Index(['Fiscal Year', 'Month Grouping', 'Month (abbrev)', 'Land Border Region',
      'State', 'Demographic', 'Citizenship', 'Title of Authority',
      'Encounter Count'],
      dtype='object')
```

```
In [374...] encounters.head()
```

Out[374...

	Fiscal Year	Month Grouping	Month (abbv)	Land Border Region	State	Demographic	Citizenship	Title of Authority	Encoun Coi
0	2020	FYTD	APR	Northern Land Border	ID	Single Adults	CANADA	Title 8	
1	2020	FYTD	APR	Northern Land Border	ME	Single Adults	CANADA	Title 42	
2	2020	FYTD	APR	Northern Land Border	ME	Single Adults	CANADA	Title 8	
3	2020	FYTD	APR	Northern Land Border	ME	Single Adults	CHINA, PEOPLES REPUBLIC OF	Title 42	
4	2020	FYTD	APR	Northern Land Border	ME	Single Adults	CHINA, PEOPLES REPUBLIC OF	Title 8	

In [375...

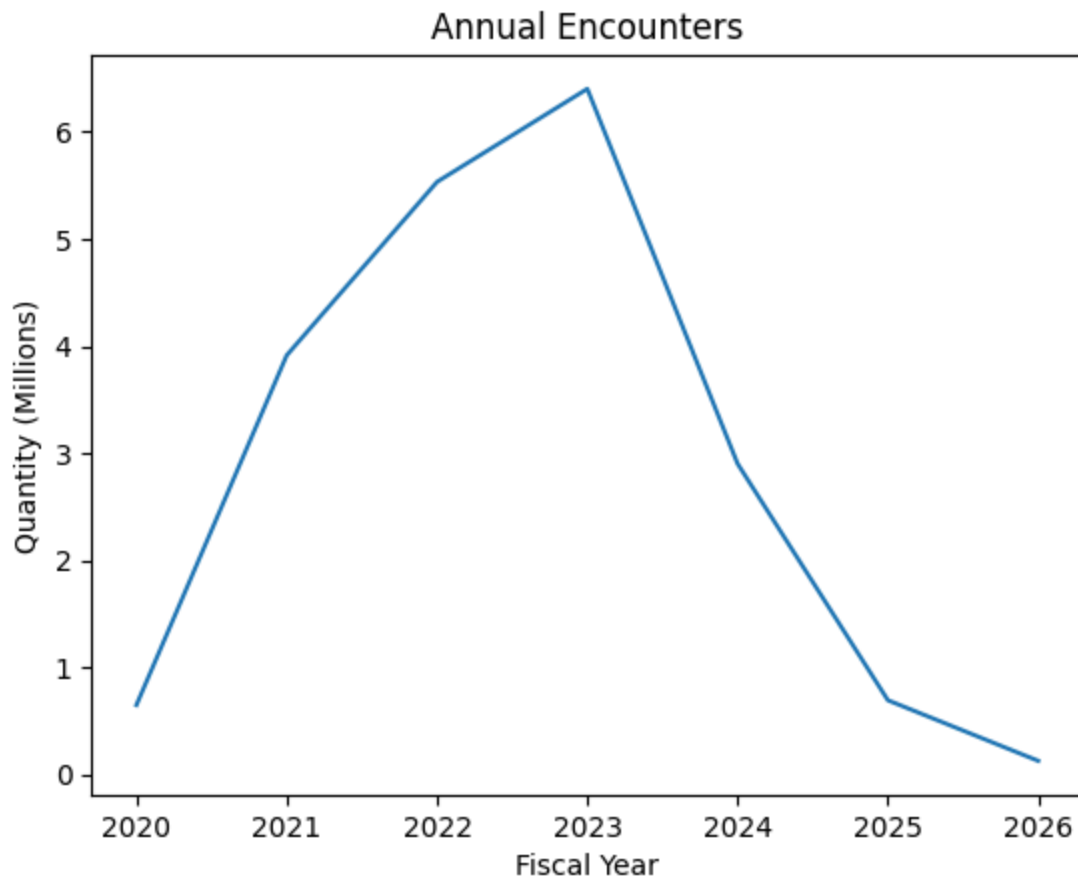
```
yearly_encounters = encounters.groupby("Fiscal Year")["Encounter Count"].sum()
```

In [376...

```
fig, ax = plt.subplots()
ax.plot(yearly_encounters.keys(), yearly_encounters.values/1000000)
ax.set_title("Annual Encounters")
ax.set_xlabel("Fiscal Year")
ax.set_ylabel("Quantity (Millions)")
```

Out[376...

```
Text(0, 0.5, 'Quantity (Millions)')
```



These establish the raw structure of the numerator and denominator.

Construction of Enforcement Intensity Ratio and Basic Statistics

To move beyond raw totals, annual encounter totals were aggregated and merged with annual DHS repatriations to construct a "repatriation to encounter" ratio:

$$Ratio = \frac{Repatriations}{Encounters}$$

This ratio represents enforcement intensity (the share of encounters that result in repatriation).

Summary Statistics for the Ratio

```
In [29]: dhs = dhs.rename(columns = {"Fiscal\nYear": "Fiscal Year"})
dhs["Fiscal Year"] = pd.to_numeric(dhs["Fiscal Year"], errors = "coerce").astype("I

dhs.head(), dhs.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Fiscal Year     16 non-null    Int64
1   Quantity        16 non-null    int64
dtypes: Int64(1), int64(1)
memory usage: 404.0 bytes

```

```

Out[29]: (   Fiscal Year  Quantity
0         2010     931170
1         2011     712850
2         2012     647530
3         2013     612030
4         2014     568250,
None)

```

```

In [30]: enc_fy = (enc1.groupby("Fiscal Year", as_index = False)["Encounter Count"]
                .sum()
                .rename(columns = {"Encounter Count": "Encounters_FY"})
                )

enc_fy.head(), enc_fy.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Fiscal Year     4 non-null    int64
1   Encounters_FY   4 non-null    int64
dtypes: int64(2)
memory usage: 196.0 bytes

```

```

Out[30]: (   Fiscal Year  Encounters_FY
0         2020         646822
1         2021        1956519
2         2022        2766582
3         2023        3201144,
None)

```

```

In [31]: fy = pd.merge(dhs.rename(columns = {"Quantity": "DHS_Quantity_FY"}),
                    enc_fy,
                    on = "Fiscal Year",
                    how = "inner"
                    )

fy["Repatriation_to_Encounter_Ratio"] = fy["DHS_Quantity_FY"] / fy["Encounters_FY"]
fy.sort_values("Fiscal Year").head(10)

```

Out[31]:

	Fiscal Year	DHS_Quantity_FY	Encounters_FY	Repatriation_to_Encounter_Ratio
0	2020	608380	646822	0.940568
1	2021	1334210	1956519	0.681931
2	2022	1467940	2766582	0.530597
3	2023	1199910	3201144	0.374838

```
In [32]: policy2 = policy.copy()
policy2["start_date"] = pd.to_datetime(policy2["start_date"], errors = "coerce")
policy2["end_date"] = pd.to_datetime(policy2["end_date"], errors = "coerce")

def fiscal_year(dt):
    # FY starts Oct 1: Oct-Dec belong to next FY
    return dt.year + (dt.month >= 10)

# create FY range per policy row
policy2["fy_start"] = policy2["start_date"].apply(fiscal_year)
policy2["fy_end"] = policy2["end_date"].apply(fiscal_year)

# explode into one row per FY
policy_fy = (policy2.loc[policy2["fy_start"].notna() & policy2["fy_end"].notna()],
             ["policy_id", "policy_name", "category", "admin", "fy_start", "fy_end"])
             .assign(Fiscal_Year_List = lambda d: d.apply(lambda r: list(range(int(
             .explode("Fiscal_Year_List")
             .rename(columns = {"Fiscal_Year_List": "Fiscal Year"}))
             )

policy_fy.head()
```

Out[32]:

	policy_id	policy_name	category	admin	fy_start	fy_end	Fiscal Year
0	ADMIN_OBAMA	Administration: Barack Obama	administration	Obama	2015	2017	2015
0	ADMIN_OBAMA	Administration: Barack Obama	administration	Obama	2015	2017	2016
0	ADMIN_OBAMA	Administration: Barack Obama	administration	Obama	2015	2017	2017
1	ADMIN_TRUMP_1	Administration: Donald Trump (1st term)	administration	Trump	2017	2021	2017
1	ADMIN_TRUMP_1	Administration: Donald Trump (1st term)	administration	Trump	2017	2021	2018

```
In [33]: fy_policy = fy.merge(policy_fy, on = "Fiscal Year", how = "left")
fy_policy.head(20)
```

Out[33]:

	Fiscal Year	DHS_Quantity_FY	Encounters_FY	Repatriation_to_Encounter_Ratio	
0	2020	608380	646822	0.940568	ADI
1	2020	608380	646822	0.940568	NE_
2	2020	608380	646822	0.940568	
3	2020	608380	646822	0.940568	
4	2020	608380	646822	0.940568	
5	2021	1334210	1956519	0.681931	ADI
6	2021	1334210	1956519	0.681931	,
7	2021	1334210	1956519	0.681931	NE_
8	2021	1334210	1956519	0.681931	
9	2021	1334210	1956519	0.681931	
10	2021	1334210	1956519	0.681931	TITLE42_UAC_
11	2021	1334210	1956519	0.681931	

	Fiscal Year	DHS_Quantity_FY	Encounters_FY	Repatriation_to_Encounter_Ratio	
12	2022	1467940	2766582	0.530597	,
13	2022	1467940	2766582	0.530597	
14	2022	1467940	2766582	0.530597	
15	2022	1467940	2766582	0.530597	TITLE42_ATTEN
16	2022	1467940	2766582	0.530597	TITLE42_UAC_
17	2022	1467940	2766582	0.530597	
18	2022	1467940	2766582	0.530597	MAYORKAS_PR
19	2023	1199910	3201144	0.374838	,

```
In [34]: dhs_annual = dhs.rename(columns = {"Fiscal\nYear": "Fiscal Year", "Quantity": "Repa
dhs_annual["Fiscal Year"] = pd.to_numeric(dhs_annual["Fiscal Year"], errors = "coer
```

```
In [35]: enc_annual = (enc1.groupby("Fiscal Year", as_index = False)["Encounter Count"]
                .sum()
                .rename(columns = {"Encounter Count": "Encounters"}))
```

```
In [36]: fy = pd.merge(dhs_annual, enc_annual, on = "Fiscal Year", how = "inner")
fy["Ratio"] = fy["Repatriations"] / fy["Encounters"]
fy.sort_values("Fiscal Year").head()
```

Out[36]:

	Fiscal Year	Repatriations	Encounters	Ratio
--	-------------	---------------	------------	-------

0	2020	608380	646822	0.940568
1	2021	1334210	1956519	0.681931
2	2022	1467940	2766582	0.530597
3	2023	1199910	3201144	0.374838

```
In [107]: fy_monthly = pd.merge(
    dhs_monthly[["Fiscal Year", "Month", "Repatriations"]],
    enc_annual[["Fiscal Year", "Encounters"]],
    on="Fiscal Year",
    how="inner"
)

fy_monthly["Ratio"] = fy_monthly["Repatriations"] / fy_monthly["Encounters"]

print("Rows in fy_annual:", len(fy_annual))
print("Rows in fy_monthly:", len(fy_monthly))
display(fy_monthly.head())
```

Rows in fy_annual: 4

Rows in fy_monthly: 48

	Fiscal Year	Month	Repatriations	Encounters	Ratio
0	2020	01 October	51360	646822	0.079404
1	2020	02 November	46120	646822	0.071302
2	2020	03 December	44920	646822	0.069447
3	2020	04 January	47060	646822	0.072756
4	2020	05 February	48730	646822	0.075338

```
In [37]: policy = pd.read_csv("policy_markers_2015_2025.csv")
policy["start_date"] = pd.to_datetime(policy["start_date"], errors = "coerce")
policy["end_date"] = pd.to_datetime(policy["end_date"], errors = "coerce")

def fiscal_year(dt):
    # FY starts Oct 1
    return dt.year + (dt.month >= 10)

policy["fy_start"] = policy["start_date"].apply(fiscal_year)
policy["fy_end"] = policy["end_date"].apply(fiscal_year)

policy_fy = (policy.dropna(subset=["fy_start", "fy_end"])
    .assign(Fiscal_Year_List = lambda d: d.apply(lambda r: list(range(int(
    .explode("Fiscal_Year_List")
    .rename(columns={"Fiscal_Year_List": "Fiscal Year"}))
    )
```

```
policy_fy = policy_fy[["Fiscal Year", "admin", "category", "policy_name", "policy_id"]]
policy_fy.head()
```

```
Out[37]:
```

	Fiscal Year	admin	category	policy_name	policy_id
0	2015	Obama	administration	Administration: Barack Obama	ADMIN_OBAMA
0	2016	Obama	administration	Administration: Barack Obama	ADMIN_OBAMA
0	2017	Obama	administration	Administration: Barack Obama	ADMIN_OBAMA
1	2017	Trump	administration	Administration: Donald Trump (1st term)	ADMIN_TRUMP_1
1	2018	Trump	administration	Administration: Donald Trump (1st term)	ADMIN_TRUMP_1

```
In [38]: fy_policy = fy.merge(policy_fy, on = "Fiscal Year", how = "left")
```

```
In [39]: repat_stats = dhs_annual["Repatriations"].agg(["count", "mean", "median", "std", "min", "max"])
repat_stats
```

```
Out[39]: count    1.600000e+01
mean      7.034331e+05
median    6.102050e+05
std       3.642389e+05
min       1.110100e+05
max       1.467940e+06
Name: Repatriations, dtype: float64
```

```
In [40]: repat_stats_df = repat_stats.to_frame(name = "Repatriations")
repat_stats_df
```

```
Out[40]:
```

	Repatriations
count	1.600000e+01
mean	7.034331e+05
median	6.102050e+05
std	3.642389e+05
min	1.110100e+05
max	1.467940e+06

```
In [41]: enc_stats = enc_annual["Encounters"].agg(["count", "mean", "median", "std", "min", "max"])
enc_stats_df = enc_stats.to_frame(name = "Encounters")
enc_stats_df
```

Out[41]:

Encounters	
count	4.000000e+00
mean	2.142767e+06
median	2.361550e+06
std	1.122771e+06
min	6.468220e+05
max	3.201144e+06

```
In [42]: ratio_stats = fy["Ratio"].agg(["count", "mean", "median", "std", "min", "max"])
ratio_stats_df = ratio_stats.to_frame(name = "Repatriation_to_Encounter_Ratio")
ratio_stats_df
```

Out[42]:

Repatriation_to_Encounter_Ratio	
count	4.000000
mean	0.631983
median	0.606264
std	0.240916
min	0.374838
max	0.940568

```
In [43]: dhs_annual["Repatriations_k"] = dhs_annual["Repatriations"] / 1_000
enc_annual["Encounters_k"] = enc_annual["Encounters"] / 1_000
```

```
In [44]: summary_table = pd.DataFrame({"Repatriations (thousands)": dhs_annual["Repatriation_k"],
                                     "Encounters (thousands)": enc_annual["Encounters_k"],
                                     "Ratio": fy["Ratio"].agg(["mean", "median", "std", "min", "max"])})

summary_table.round(3)
```

Out[44]:

	Repatriations (thousands)	Encounters (thousands)	Ratio
mean	703.433	2142.767	0.632
median	610.205	2361.550	0.606
std	364.239	1122.771	0.241
min	111.010	646.822	0.375
max	1467.940	3201.144	0.941

The relatively large standard deviation indicates meaningful variability across fiscal years. Most notably, the ratio declines steadily even as raw encounter volume increases

dramatically.

This confirms that enforcement output does not scale proportionally with inflow pressure.

```
In [88]: enc_annual = (enc1.groupby("Fiscal Year", as_index = False)["Encounter Count"]
        .sum()
        .rename(columns = {"Encounter Count": "Encounters"}))

print(enc_annual["Fiscal Year"].unique())
enc_annual.head()
```

[2020 2021 2022 2023]

Out[88]:

	Fiscal Year	Encounters
--	-------------	------------

0	2020	646822
1	2021	1956519
2	2022	2766582
3	2023	3201144

```
In [90]: dhs_month = pd.read_excel("KHSR Repatriations fy25m11.xlsx",
        sheet_name = "Monthly Repatriations")

# clean column names
dhs_month.columns = (dhs_month.columns.astype(str)
        .str.replace("\n", " ", regex = False)
        .str.replace(r"\s+", " ", regex = True)
        .str.strip()
        )

print(dhs_month.columns)
dhs_month.head()
```

Index(['Fiscal Year', 'Month', 'Quantity'], dtype='object')

Out[90]:

	Fiscal Year	Month	Quantity
--	-------------	-------	----------

0	2010	01 October	79200
1	2010	02 November	69680
2	2010	03 December	63560
3	2010	04 January	69200
4	2010	05 February	75750

```
In [91]: # find likely columns
fy_col = [c for c in dhs_month.columns if "Fiscal" in c][0]
qty_col = [c for c in dhs_month.columns if "Quantity" in c][0]
month_col = [c for c in dhs_month.columns if "Month" in c][0]

dhs_monthly = dhs_month.rename(columns = {fy_col: "Fiscal Year",
```

```

        month_col: "Month",
        qty_col: "Repatriations"
    }).copy()

# make sure FY numeric
dhs_monthly["Fiscal Year"] = pd.to_numeric(dhs_monthly["Fiscal Year"], errors = "co

# keep only rows with actual data
dhs_monthly = dhs_monthly.dropna(subset=["Fiscal Year", "Month", "Repatriations"])

print("DHS monthly rows:", len(dhs_monthly))
dhs_monthly.head()

```

DHS monthly rows: 182

Out[91]:

	Fiscal Year	Month	Repatriations
0	2010	01 October	79200
1	2010	02 November	69680
2	2010	03 December	63560
3	2010	04 January	69200
4	2010	05 February	75750

In [92]:

```

fy_monthly = pd.merge(
    dhs_monthly[["Fiscal Year", "Month", "Repatriations"]],
    enc_annual[["Fiscal Year", "Encounters"]],
    on="Fiscal Year",
    how="inner"
)

fy_monthly["Ratio"] = fy_monthly["Repatriations"] / fy_monthly["Encounters"]

print("fy_monthly rows:", len(fy_monthly))
print(fy_monthly["Fiscal Year"].value_counts())
fy_monthly.head()

```

fy_monthly rows: 48

Fiscal Year

2020 12

2021 12

2022 12

2023 12

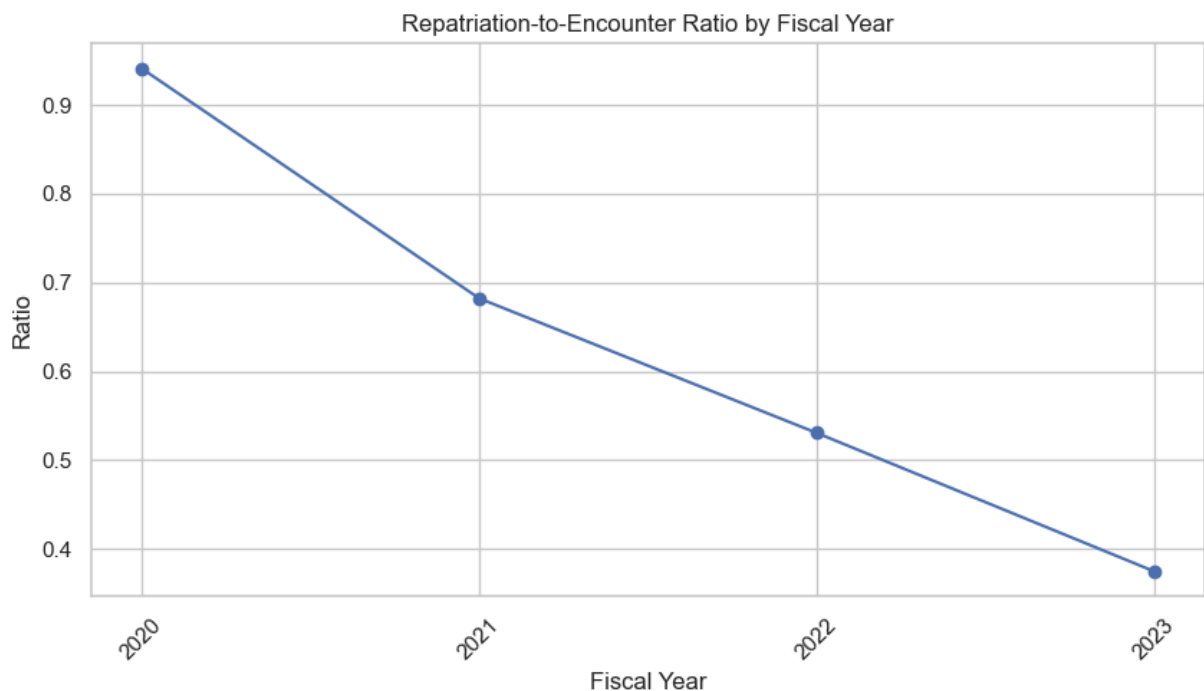
Name: count, dtype: int64

Out[92]:

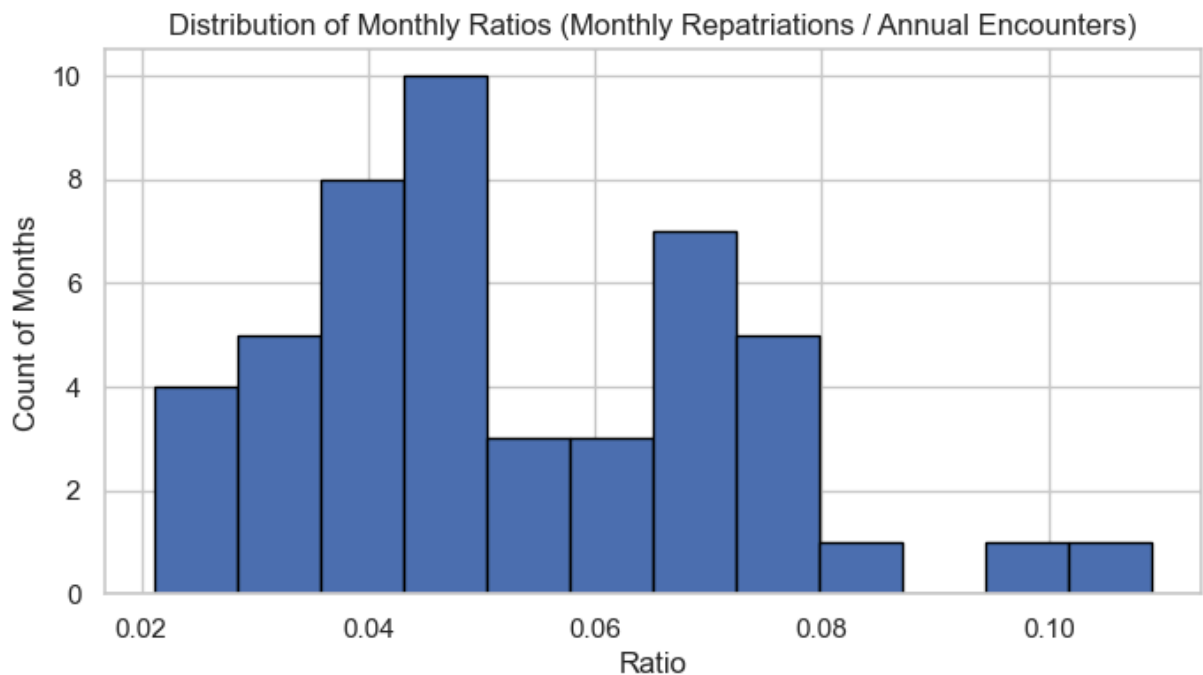
	Fiscal Year	Month	Repatriations	Encounters	Ratio
0	2020	01 October	51360	646822	0.079404
1	2020	02 November	46120	646822	0.071302
2	2020	03 December	44920	646822	0.069447
3	2020	04 January	47060	646822	0.072756
4	2020	05 February	48730	646822	0.075338

```
In [97]: fy_sorted = fy.sort_values("Fiscal Year")
```

```
plt.figure(figsize = (10,5))
plt.plot(fy_sorted["Fiscal Year"], fy_sorted["Ratio"], marker = "o")
plt.title("Repatriation-to-Encounter Ratio by Fiscal Year")
plt.xlabel("Fiscal Year")
plt.ylabel("Ratio")
plt.xticks(fy_sorted["Fiscal Year"], rotation = 45)
plt.show()
```



```
In [96]: plt.figure(figsize = (8,4))
plt.hist(fy_monthly["Ratio"].dropna(), bins = 12, edgecolor = "black")
plt.title("Distribution of Monthly Ratios (Monthly Repatriations / Annual Encounter")
plt.xlabel("Ratio")
plt.ylabel("Count of Months")
plt.show()
```



```
In [58]: print(fy[["Repatriations", "Encounters", "Ratio"]])
```

	Repatriations	Encounters	Ratio
0	608380	646822	0.940568
1	1334210	1956519	0.681931
2	1467940	2766582	0.530597
3	1199910	3201144	0.374838

Structural Patterns and Regime Differences

The Ratio by Fiscal Year line plot shows a clear downward trend starting around FY2020. Even though repatriations increase at first, they do not keep up with the rapid growth in encounters. As encounters surge, the ratio steadily declines, meaning a smaller share of encounters results in repatriation over time. This creates a noticeable gap between inflow pressure (encounters) and enforcement output (repatriations). The visual makes this pattern much easier to see than looking at raw numbers alone.

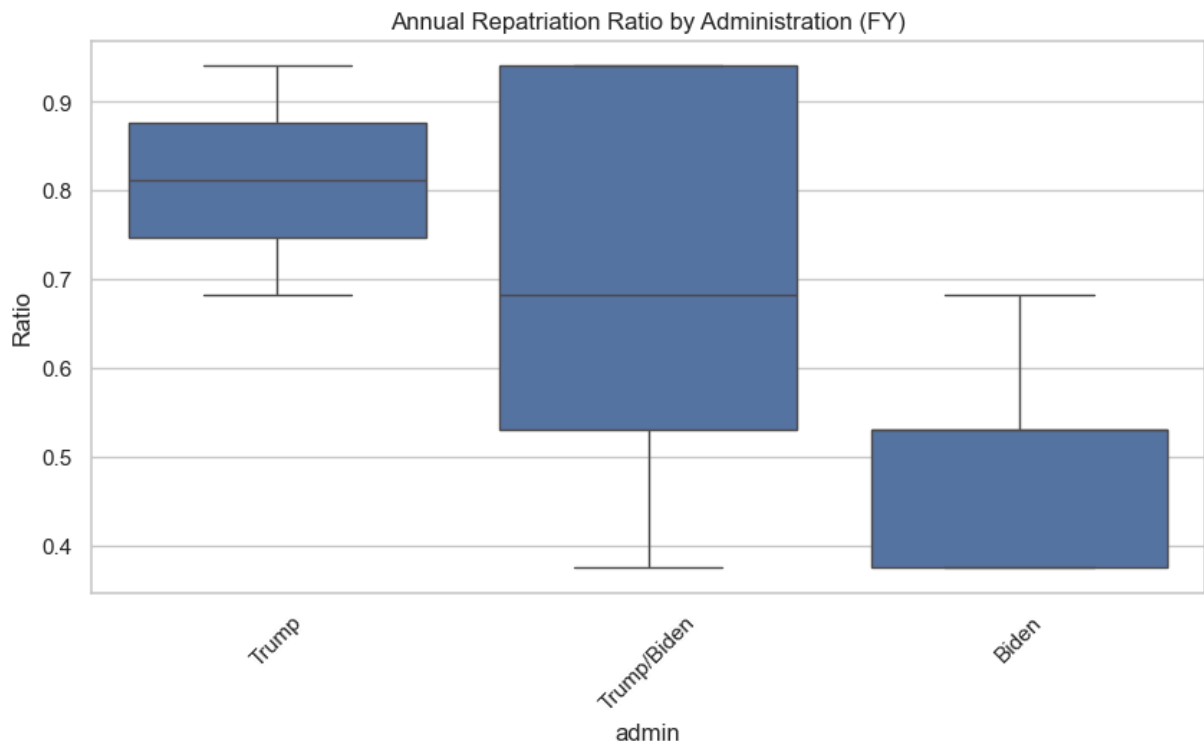
To explore whether these changes might be connected to broader institutional shifts, a boxplot of the ratio by administration was created. The boxplot shows that enforcement intensity is not evenly distributed across administrations. Instead, the ratio varies in level and spread depending on the time period. This suggests that enforcement outcomes may be influenced by broader regime conditions, not just by how many people are encountered at the border.

The Encounters by Title of Authority visualization adds another layer to this analysis. It shows how the use of Title 42 and Title 8 changes over time, especially during peak encounter years. Since Title 42 represents a major legal mechanism during the public health emergency period, shifts in authority likely affect how encounters are processed. Seeing these authority changes alongside the declining ratio provides additional context for why enforcement intensity may be shifting.

Overall, these visuals support the project's main objective, where instead of focusing only on

raw totals, the analysis looks for structural differences in enforcement intensity that align with broader policy regimes. The patterns in the ratio line plot, administration boxplot, and authority breakdown together suggest that enforcement dynamics change across institutional environments rather than remaining constant over time.

```
In [71]: plt.figure(figsize = (10,5))
sns.boxplot(data = fy_policy, x = "admin", y = "Ratio")
plt.title("Annual Repatriation Ratio by Administration (FY)")
plt.xticks(rotation = 45)
plt.show()
```



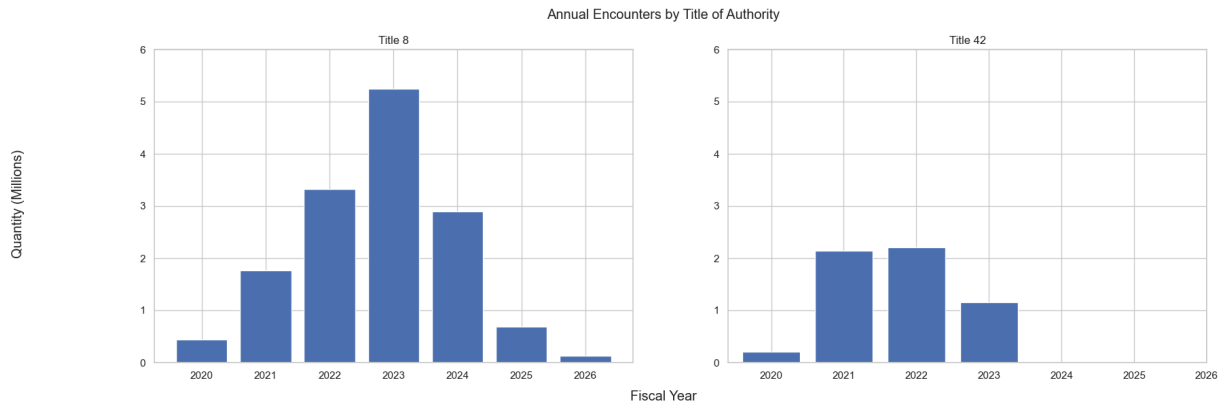
```
In [106... formatter = matplotlib.ticker.StrMethodFormatter("{x:.0f}")
figure = plt.figure(figsize=(20, 6))

authorities = encounters["Title of Authority"].unique()

for i, authority in enumerate(authorities):
    axes = figure.add_subplot(1, len(authorities), i+1)
    authority_encounters = encounters[encounters["Title of Authority"] == authority]
    axes.bar(authority_encounters.keys(), authority_encounters.values/1000000)
    axes.set_title(f"{authority}")
    axes.set_ylim((0,6))
    axes.set_xticks(range(2020, 2027))
    plt.gca().xaxis.set_major_formatter(formatter)

figure.suptitle("Annual Encounters by Title of Authority")
figure.supylabel("Quantity (Millions)")
figure.supxlabel("Fiscal Year")
```

Out[106... Text(0.5, 0.01, 'Fiscal Year')



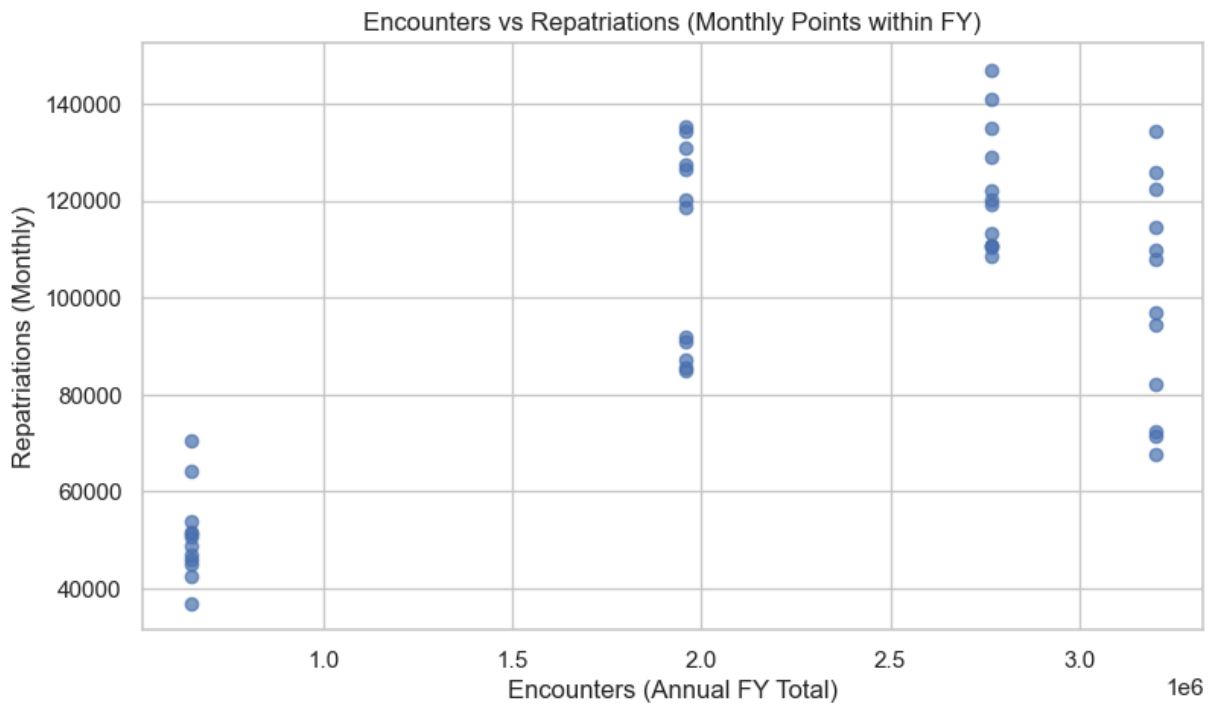
Scaling Behavior and Capacity Constraints

To better understand how encounter volume relates to enforcement outcomes, a scatter plot comparing Encounters and Repatriations was created. This visualization helps show whether removals increase at the same rate as inflow pressure or if the relationship changes as encounters rise. The scatter plot shows that repatriations do generally increase as encounters increase, but the relationship is not proportional. At lower encounter levels, repatriations appear more tightly clustered. However, as encounters reach higher levels, the spread in repatriation totals becomes much wider. This means enforcement output becomes less predictable during peak inflow periods. In other words, the system does not seem to operate at a constant processing rate. Instead of scaling evenly with encounter growth, repatriations fluctuate more as pressure increases.

This pattern suggests that enforcement intensity may decline during periods of rapid encounter growth. When encounters surge, the share that results in repatriation appears to fall, which could reflect operational capacity limits, legal constraints, or broader institutional shifts. Rather than simply reflecting higher volume, the relationship between encounters and removals seems sensitive to structural conditions.

This directly connects to one of the additional research questions outlined in the project proposal, which is whether periods of rapid encounter growth correspond with declining repatriation ratios, potentially indicating operational capacity constraints. The scatter plot provides early exploratory evidence that such a relationship may exist and warrants further structural analysis.

```
In [95]: plt.figure(figsize = (9,5))
plt.scatter(fy_monthly["Encounters"], fy_monthly["Repatriations"], alpha = 0.7)
plt.xlabel("Encounters (Annual FY Total)")
plt.ylabel("Repatriations (Monthly)")
plt.title("Encounters vs Repatriations (Monthly Points within FY)")
plt.show()
```



***Refinement of Research Questions**

The initial question asked whether changes in repatriation ratios are associated with identifiable federal policy regime shifts. After examining the visuals and summary statistics, it is clear that the ratio does not remain stable over time. Instead, it shows noticeable structural movement beginning around FY2020, which suggests that something broader than random fluctuation may be occurring.

The ratio line plot, boxplot by administration, and scatter plot together show that enforcement intensity is not constant and does not scale proportionally with encounter volume. As encounters increase sharply, the share resulting in repatriation declines. This suggests that enforcement output may be influenced by institutional conditions rather than simply by inflow pressure alone.

Based on these findings, the research focus shifts slightly. Instead of only asking whether regime shifts are associated with changes in the ratio, the analysis now considers more specific questions:

- Does enforcement intensity decline during high volume encounter periods?
- Do legal regime shifts, such as the implementation of Title 42, alter the relationship between encounters and removals?
- Are observable structural breaks in the ratio aligned with administration transitions?
- Does enforcement variability increase during emergency or public health periods?

The EDA results suggest that enforcement dynamics are sensitive to regime context.

Rather than comparing raw counts alone, the next stage of analysis should focus on structural classification of policy periods and testing for regime aligned shifts in enforcement intensity. This approach remains consistent with the project's objective of identifying structural differences across enforcement environments

